

Vitor Augusto Hertel Rovaris

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EDUCATION

Purdue University

B.S. in Computer Science, Concentration in Systems Software and Security

West Lafayette, IN

Aug. 2024 – May 2028

GPA: 3.84/4.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Systems Programming, Computer Security.

EXPERIENCE

Purdue University

Undergraduate Research Assistant — AutoIC Lab

West Lafayette, IN

Aug. 2026 – Present

- Awarded a Purdue Office of Undergraduate Research Scholarship for the 2026–2027 academic year to research AI, natural language processing, and Building Information Modeling applied to construction automation, in a lab funded by the National Science Foundation.

Undergraduate Teaching Assistant — CS 18000

Aug. 2026 – Present

- Lead weekly lab sections of 25 students, hold office hours, and proctor exams for CS 18000: Problem Solving and Object-Oriented Programming, Purdue's introductory computer science course taken by 761 students this term, coaching students through algorithm design, debugging, and object-oriented design in Java.

BTG Pactual

Software Engineering Intern — Energy Desk

São Paulo, Brazil

June 2026 – Aug. 2026

- Built an event-driven integration between the Energy Desk's backend and the bank's core finance system using Python, RabbitMQ, and Celery, deployed on Kubernetes with PostgreSQL.
- Automated tracking of debit notes across 3 internal systems, processing 371 notes/month representing over \$140M in monthly financial flows, with individual notes ranging from \$0.01 to \$236M.
- Closed a \$50M discrepancy between the bank's managerial and accounting reports; delivered the system into UAT for business validation ahead of production rollout.

Software Engineering Intern

June 2025 – Aug. 2025

- Designed a serverless AI pipeline (AWS Lambda, SQS) for automated contract verification, processing 20,000+ signatures in hours; presented 80% efficiency gains to executives, driving enterprise-wide AI adoption.

BRASA – Brazilian Student Association

Technology Director

Remote

Sept. 2025 – Aug. 2026

- Defined and executed the 2025–2026 technical roadmap across web, mobile, and data platforms serving 20,000+ members; managed 13 people across Product & UX, Software Engineering, and Data teams.
- Led architecture reviews and set technical standards across multiple products; structured the migration of 12 years of legacy data into the platform's unified PostgreSQL database.

Software Engineering Manager

May 2025 – Sept. 2025

- Managed 4 software engineers; established a structured development lifecycle with design reviews, documentation standards, and security checks, raising Net Promoter Score by 120+ points.

Software Engineering Analyst

Oct. 2024 – May 2025

- Built a React/Node.js portal with PostgreSQL, REST APIs, and role-based authentication; engineered a React Native app reaching 90%+ adoption across the 20,000+ member network.

PROJECTS

Medical Shift Scheduling Platform | *Next.js, TypeScript, Supabase, Python, OR-Tools*

June 2026 – Sept. 2026

- Developed a full-stack scheduling platform (Next.js, TypeScript, Supabase/PostgreSQL) that assigns a 25-doctor anesthesiology group to operating rooms and on-call shifts, backed by a Python/FastAPI microservice using Google OR-Tools CP-SAT to encode rest rules and room workloads.

Hermes — Secure Linux Sandbox | *C, Seccomp-BPF, cgroups v2, FastAPI*

Apr. 2026 – July 2026

- Built a code execution engine in C that runs untrusted binaries in isolated PID, network, and mount namespaces with a chroot jail, seccomp-BPF syscall whitelisting, and cgroups v2 memory caps; exposed through a FastAPI service and validated against 7 attack scenarios.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, SQL, Assembly (ARM, x86)

Web & Backend: React, React Native, Next.js, Node.js, Django REST Framework

Cloud & Infrastructure: AWS (Lambda, SQS, EC2), Kubernetes, Docker, RabbitMQ, Celery, PostgreSQL, Linux/UNIX, Git